

Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Modul 5: VLAN, Trunk, dan OSPF Multivendor

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1 Langkah Percobaan

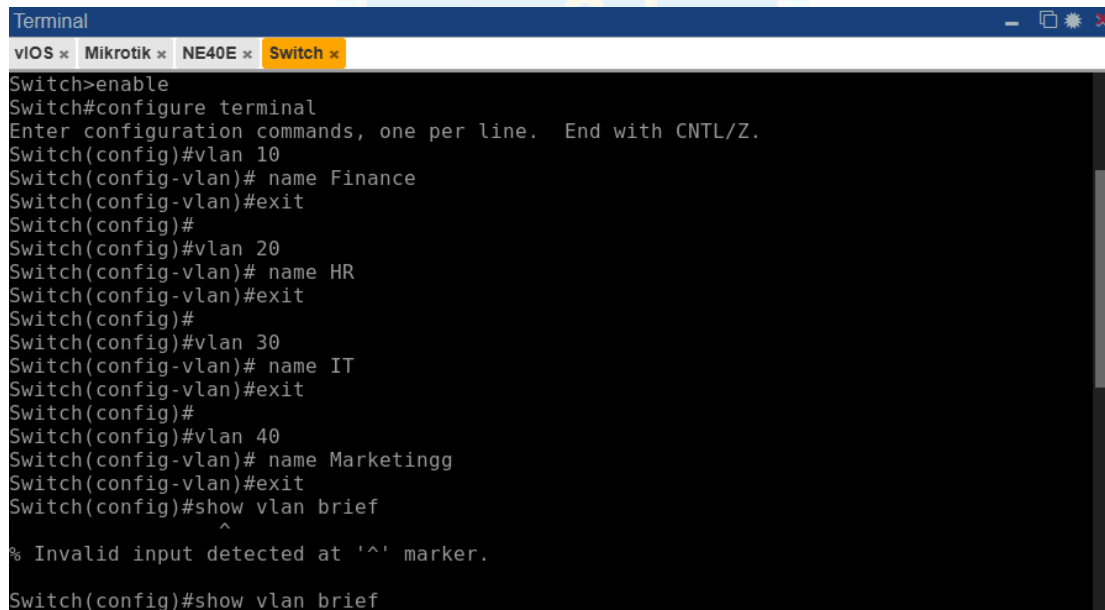
1.1 Konfigurasi Cisco Switch

1. Masuk ke mode privileged dan mode konfigurasi Cisco Switch.

```
enable
configure terminal
```

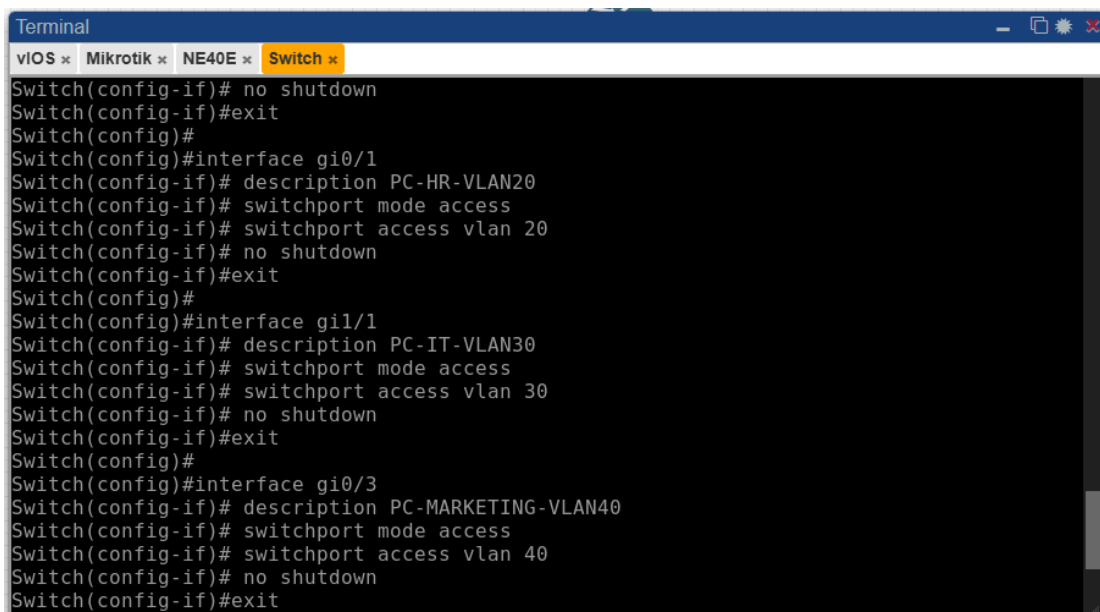
2. Membuat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing), setelah itu setting port.

```
vlan 10
  name Finance
vlan 20
  name HR
vlan 30
  name IT
vlan 40
  name Marketing
```



Gambar 1: Pembuatan VLAN

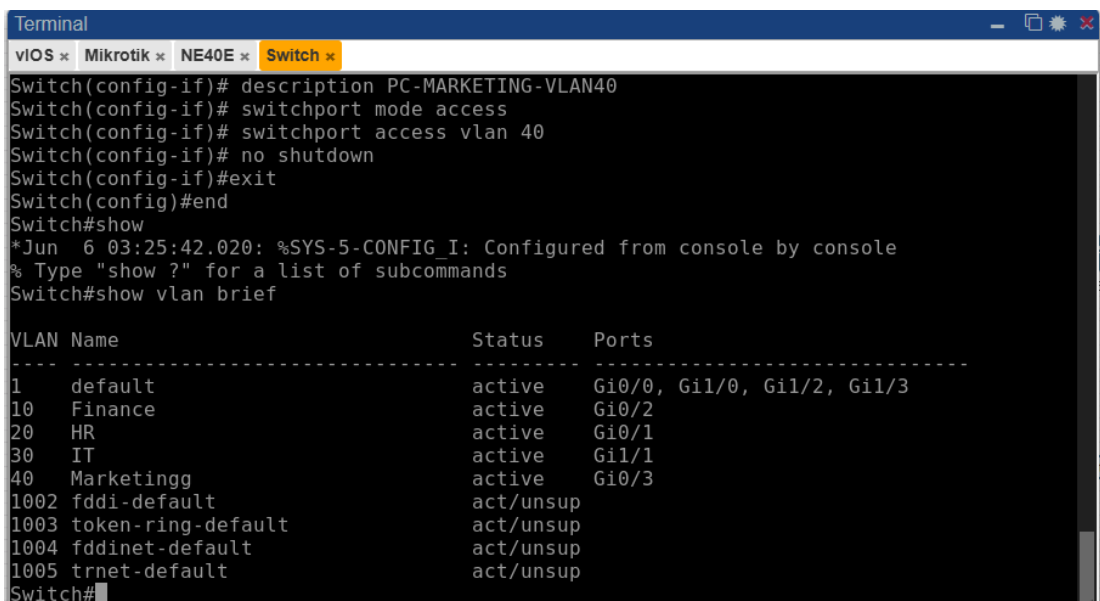
```
interface gi0/2
  switchport mode access
  switchport access vlan 10
interface gi0/1
  switchport mode access
  switchport access vlan 20
interface gi1/1
  switchport mode access
  switchport access vlan 30
interface gi0/3
  switchport mode access
  switchport access vlan 40
```



```
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/1
Switch(config-if)# description PC-HR-VLAN20
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi1/1
Switch(config-if)# description PC-IT-VLAN30
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gi0/3
Switch(config-if)# description PC-MARKETING-VLAN40
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
```

Gambar 2: Konfigurasi access port

3. verifikasi VLAN dan Access port menggunakan command `show vlan brief`.



```
Switch(config-if)# description PC-MARKETING-VLAN40
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#show
*Jun  6 03:25:42.020: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi1/0, Gi1/2, Gi1/3
10	Finance	active	Gi0/2
20	HR	active	Gi0/1
30	IT	active	Gi1/1
40	Marketingg	active	Gi0/3
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#
```

Gambar 3: Hasil verifikasi VLAN dan access port pada Cisco Switch.

4. Mengatur interface Gi0/0 yang menuju Cisco Router sebagai trunk IEEE 802.1Q dan mengizinkan VLAN 10, 20, 30, dan 40.

```
interface gi0/0
description TRUNK-TO-CISCO-ROUTER
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
```

```
Terminal
vIOS x Mikrotik x NE40E x Switch x

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Switch#conf t
Switch(config)#
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-T0-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30,40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
Switch(config)#exit
Switch# description TRUNK-T0-CISCO-ROUTER
```

Gambar 4: konfigurasi trunk Cisco Switch.

5. Verifikasi trunk dengan command `show interfaces trunk`.

```
Terminal
vIOS x Mikrotik x NE40E x Switch x

Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
Switch(config)#exit
Switch# description TRUNK-T0-CISCO-ROUTER
*Jun 6 03:28:02.766: %SYS-5-CONFIG_I:
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

Gambar 5: Hasil verifikasi trunk pada interface Gi0/0.

1.2 Konfigurasi Cisco Router

1. Mengaktifkan interface Gi0/2 sebagai interface fisik trunk tanpa IP, lalu membuat subinterface sebagai gateway setiap VLAN, Menggunakan command ini :

```
interface gi0/2
description TRUNK-T0-CISCO-SWITCH
no ip address
no shutdown

interface gi0/2.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

```

interface gi0/2.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
interface gi0/2.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
interface gi0/2.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0

```

2. Verifikasi subinterface dengan command `show ip interface brief`.

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x
Router(config)#show ip interface brief
^
% Invalid input detected at '^' marker.
Router(config)#exit
Router#conf t
*Jun  6 03:36:43.013: %SYS-5-CONFIG_I: Configured from console by c
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down  down
GigabitEthernet0/1       unassigned      YES unset  administratively down  down
GigabitEthernet0/2       unassigned      YES unset  up                    up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up                    up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3       unassigned      YES unset  administratively down  down
Router#

```

Gambar 6: Informasi semua interface berstatus up

3. Mengaktifkan DHCP Server untuk VLAN 10 dan VLAN 20. Alamat 192.168.10.1–192.168.10.10 dan 192.168.20.1–192.168.20.10 dikecualikan agar tidak dibagikan kepada klien.

```

ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool VLAN10-FINANCE
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 8.8.8.8

ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp pool VLAN20-HR
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 8.8.8.8

```

4. Meminta alamat DHCP pada PC VLAN 10 dan VLAN 20, sedangkan PC VLAN 30 dan VLAN 40 diberi IP statis.

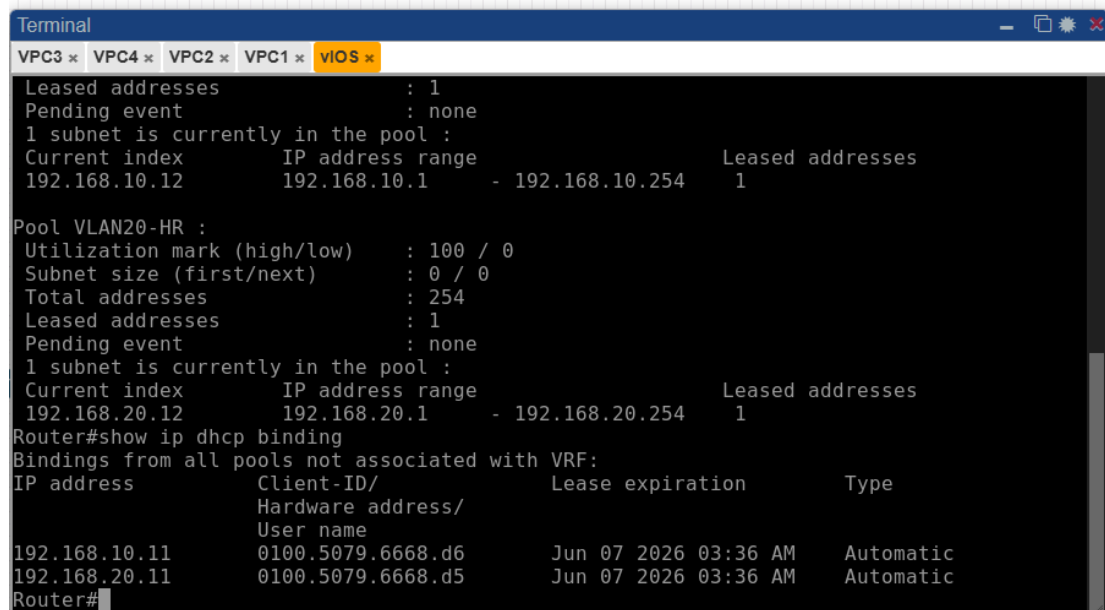
```

PC VLAN 10 : ip dhcp
PC VLAN 20 : ip dhcp
PC VLAN 30 : ip 192.168.30.10/24 192.168.30.1

```

PC VLAN 40 : ip 192.168.40.10/24 192.168.40.1

5. Memeriksa pool dan binding DHCP.



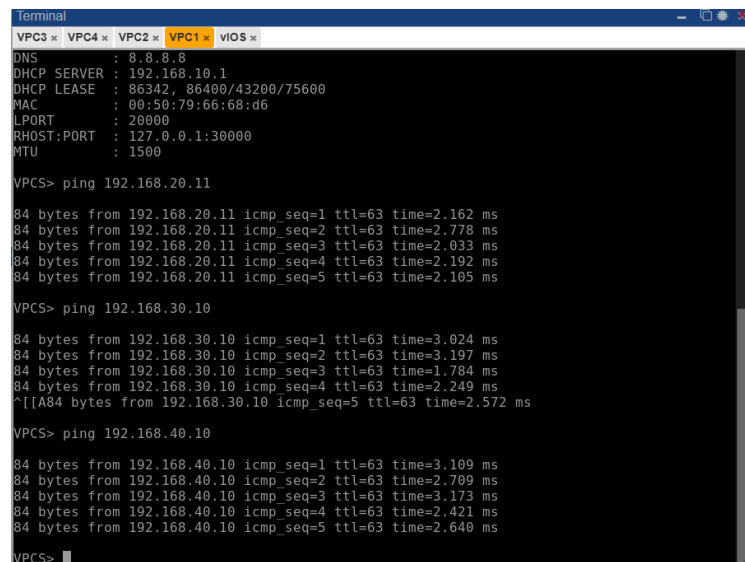
```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
User name
192.168.10.11 0100.5079.6668.d6 Jun 07 2026 03:36 AM Automatic
192.168.20.11 0100.5079.6668.d5 Jun 07 2026 03:36 AM Automatic
Router#
```

Gambar 7: Verifikasi DHCP pool dan DHCP binding.

6. Menguji koneksi ke gateway dan komunikasi antar-VLAN.



```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x
DNS : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE : 86342, 86400/43200/75600
MAC : 00:50:79:66:68:d6
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.162 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.033 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.192 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.197 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
^[[A84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.109 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms

VPCS>
```

Gambar 8: Pengujian ping antar-VLAN dari VPCS.

1.3 Konfigurasi Huawei dan Link Antar-router

1. Mengatur LAN Huawei pada Ethernet1/0/2 menggunakan gateway 192.168.50.1/24. PC LAN Huawei dikonfigurasi dengan IP 192.168.50.10/24 dan gateway 192.168.50.1 menggunakan command :

```
system-view
interface Ethernet1/0/2
```

```

description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit
commit

```

(a) Konfigurasi LAN Huawei

(b) IP statis PC LAN Huawei

Gambar 9: Dokumentasi konfigurasi LAN pada sisi Huawei.

2. Mengatur empat network point-to-point. Dua link Cisco–Huawei menggunakan 10.10.10.1, 10.10.11.1, dan 10.10.30.1; Huawei menggunakan 10.10.10.2, 10.10.11.2, dan 10.10.20.1; sedangkan MikroTik menggunakan 10.10.20.2 dan 10.10.30.2.

(a) Dua link Cisco–Huawei

(b) Dua link Huawei–Cisco

(c) Link Cisco–MikroTik

(d) Link MikroTik–Huawei dan MikroTik–Cisco

Gambar 10: Dokumentasi Konfigurasi.

3. Verifikasi alamat dan status interface pada ketiga router.

```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
*Jun 6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip address 10.10.30.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#show
*Jun 6 03:46:29.797: %SYS-5-CONFIG I: Configured from console by console
% Type "show ?" for a list of subcommands
Router#show ip interface brief
^
% Invalid input detected at '^' marker.
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#display ip interface brief
^
```

(a) Cisco Router

```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7
Interface IP Address/Mask Physical Protocol VPN
Ethernet1/0/0 10.10.10.2/30 up up --
Ethernet1/0/0.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/1 unassigned up down --
Ethernet1/0/1.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/2 192.168.50.1/24 up up --
Ethernet1/0/2.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/3 10.10.20.1/30 up up --
Ethernet1/0/3.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/4 10.10.11.2/30 up up --
Ethernet1/0/4.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/5 unassigned up down --
Ethernet1/0/5.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/6 unassigned up down --
Ethernet1/0/6.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/7 unassigned up down --
Ethernet1/0/7.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/8 unassigned up down --
Ethernet1/0/8.4094 128.1.138.137/16 up up 13vpn
Ethernet1/0/9 unassigned up down --
Ethernet1/0/9.4094 128.1.138.137/16 up up 13vpn
GigabitEthernet0/0/0 unassigned up down --
---- More ----
```

(b) Huawei Router

Gambar 11: Verifikasi interface Cisco dan Huawei.

```
Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

[admin@MikroTik] >> /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >> /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
[admin@MikroTik] >> ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; T0-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; T0-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >>
```

Gambar 12: Verifikasi IP address pada MikroTik.

4. Melakukan ping antar-router untuk memastikan seluruh link langsung dapat digunakan sebelum OSPF diaktifkan.

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

Router#display ip interface brief
^
% Invalid input detected at '^' marker.

Router#display interface brief
^
% Invalid input detected at '^' marker.

Router#display ip interface brief
^
% Invalid input detected at '^' marker.

Router#display ip interface brief
^
% Invalid input detected at '^' marker.

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

(a) Ping dari Cisco

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss

```

(b) Ping dari Huawei

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x

sent=20 received=20 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=7ms
SEQ HOST                SIZE TTL TIME STATUS
20 10.10.20.1            56 255 1ms
21 10.10.20.1            56 255 1ms
22 10.10.20.1            56 255 1ms
23 10.10.20.1            56 255 2ms
24 10.10.20.1            56 255 1ms
25 10.10.20.1            56 255 1ms
26 10.10.20.1            56 255 3ms
27 10.10.20.1            56 255 1ms
28 10.10.20.1            56 255 1ms
29 10.10.20.1            56 255 1ms
30 10.10.20.1            56 255 0ms
31 10.10.20.1            56 255 1ms
32 10.10.20.1            56 255 1ms
sent=33 received=33 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=7ms

[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                SIZE TTL TIME STATUS
0 10.10.20.1            56 255 2ms
1 10.10.20.1            56 255 2ms
2 10.10.20.1            56 255 1ms
3 10.10.20.1            56 255 2ms
sent=4 received=4 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                SIZE TTL TIME STATUS
0 10.10.30.1            56 255 0ms
1 10.10.30.1            56 255 2ms
2 10.10.30.1            56 255 2ms
3 10.10.30.1            56 255 5ms
sent=4 received=4 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=5ms

[admin@MikroTik] >

```

Gambar 13: Ping antar router

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengaktifkan OSPF area 0 pada Cisco Router dengan router-id 1.1.1.1. Network yang diiklankan mencakup seluruh VLAN serta link menuju Huawei dan MikroTik. Cost link Cisco–Huawei diatur 10, sedangkan link Cisco–MikroTik diatur 100.

```

router ospf 1
router-id 1.1.1.1
network 10.10.10.0 0.0.0.3 area 0
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0

```

```

interface gi0/3
  ip ospf cost 10
interface gi0/0
  ip ospf cost 10
interface gi0/1
  ip ospf cost 100

```

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
Router(config-router)# network 10.10.11.0 0.0.0.3 area 0
Router(config-router)# network 10.10.30.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)# description LINK1-T0-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-T0-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/1
Router(config-if)# description LINK-T0-MIKROTIK
Router(config-if)# ip ospf cost 100
Router(config-if)#exit
Router(config)#

```

Gambar 14: Konfigurasi OSPF dan cost Cisco

2. Mengaktifkan OSPF pada Huawei Router dengan router-id 2.2.2.2. Huawei mengiklankan dua link ke Cisco, link ke MikroTik, dan LAN 192.168.50.0/24. Cost link ke Cisco diatur 10 dan cost link ke MikroTik diatur 100.

```

ospf 1 router-id 2.2.2.2
  area 0.0.0.0
    network 10.10.10.0 0.0.0.3
    network 10.10.11.0 0.0.0.3
    network 10.10.20.0 0.0.0.3
    network 192.168.50.0 0.0.0.255

interface Ethernet1/0/0
  ospf cost 10
interface Ethernet1/0/4
  ospf cost 10
interface Ethernet1/0/3
  ospf cost 100
commit

```

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
[~HUAWEI]system -v~w
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]system -view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
^
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1] area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0] network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0] quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0] description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0] ospf cost 10
[*HUAWEI-Ethernet1/0/0] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4] ospf cost 10
[*HUAWEI-Ethernet1/0/4] quit
[*HUAWEI]
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 15: Konfigurasi OSPF dan cost Huawei

3. Mengaktifkan OSPF pada MikroTik dengan router-id 3.3.3.3. MikroTik mengiklankan network 10.10.20.0/30 dan 10.10.30.0/30 sebagai jalur cadangan dengan cost 100.

```

/routing ospf instance set default router-id=3.3.3.3
/routing ospf network add network=10.10.20.0/30 area=backbone
/routing ospf network add network=10.10.30.0/30 area=backbone
/routing ospf interface add interface=ether1 cost=100
/routing ospf interface add interface=ether2 cost=100

```

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
syntax error (line 1 column 154)
[admin@MikroTik] >> /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] >> /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] >> routing ospf neighbor print

[admin@MikroTik] > /routing ospf neighbor print

[admin@MikroTik] > /routing ospf interface add network=10.10.20.0/30 area=backbone
syntax error (line 1 column 37)
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone

[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=7s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3s
[admin@MikroTik] >

```

Gambar 16: Konfigurasi OSPF multivendor.

4. Memverifikasi adjacency OSPF. Cisco membentuk tiga neighbor, yaitu dua adjacency ke Huawei dan satu adjacency ke MikroTik.

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
Distance: (default is 4)
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.0.0/24 area 0
    10.10.11.0/24 area 0
    10.10.30.0/24 area 0
    192.168.10.0/24 area 0
    192.168.20.0/24 area 0
    192.168.30.0/24 area 0
    192.168.40.0/24 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

Router#
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 fro
LOADING to FULL, Loading Done
*Jun  6 03:59:00.409: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
Router#show ip ospf neighbor

```

(a) Neighbor OSPF Cisco

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 28 sec
Retrans timer interval: 5
Neighbor is up for 00h00m14s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 39 sec
Retrans timer interval: 5
Neighbor is up for 00h01m53s
Neighbor Up Time : 2026-06-06 04:26:06
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 28 sec
Retrans timer interval: 5
Neighbor is up for 00h00m14s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]

```

(b) Peer OSPF Huawei

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x VIOS x NE40E x Mikrotik x VPC0 x
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 Ado 10.10.20.0/30 10.10.20.2 ether1 0
3 Ado 10.10.30.0/30 10.10.30.2 ether2 0
4 Ado 192.168.10.0/24 10.10.30.1 110
5 Ado 192.168.20.0/24 10.10.30.1 110
6 Ado 192.168.30.0/24 10.10.30.1 110
7 Ado 192.168.40.0/24 10.10.30.1 110
8 Ado 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >> /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m51s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m47s
[admin@MikroTik] >>

```

Gambar 17: Hasil pembentukan neighbor OSPF multivendor.

- Memeriksa tabel routing OSPF. Cisco memperoleh route menuju LAN Huawei, sedangkan Huawei dan MikroTik memperoleh route menuju network VLAN.

```

Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/DR 00:00:38 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:38 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:37 10.10.10.2 GigabitEthernet0/3

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:09:25 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:09:25 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:09:35 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
  10.10.20.0/30 [110/110] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
  [110/110] via 10.10.10.2, 00:09:45, GigabitEthernet0/3
  192.168.50.0/24 [110/11] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
  [110/11] via 10.10.10.2, 00:09:45, GigabitEthernet0/3

Router#dis

```

(a) Route OSPF Cisco

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Public_Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 110 D 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]

```

(b) Route OSPF Huawei

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x vIOS x NE40E x Mikrotik x VPC0 x

4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110

[admin@MikroTik] >> /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m51s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m47s

[admin@MikroTik] >> routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=8m49s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=8m45s

[admin@MikroTik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.30.1 110
1 ADo 10.10.11.0/30 10.10.30.1 110
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110

[admin@MikroTik] >>

```

Gambar 18: Tabel routing OSPF pada perangkat multivendor.

6. Menguji koneksi end-to-end dari VLAN Cisco menuju LAN Huawei 192.168.50.10.

```

Terminal
VPC3 x VPC4 x VPC2 x VPC1 x vIOS x NE40E x Mikrotik x VPC0 x

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.162 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.033 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.192 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.197 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
^[[A84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.109 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.522 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.330 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.060 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.090 ms

VPCS>

```

Gambar 19: Pengujian ping end-to-end dari VLAN menuju LAN Huawei.

1.5 Pengujian Failover OSPF

1. Pada kondisi normal, kedua link Cisco–Huawei aktif. Hasil `trace 192.168.50.10` melewati gateway VLAN, Cisco Router, lalu langsung menuju Huawei melalui link utama dengan cost 10.

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    1.863 ms  1.899 ms  1.659 ms
 2  10.10.10.2     3.725 ms  1.951 ms  1.827 ms
 3  *192.168.50.10  2.342 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 20: Kondisi Awal

2. Salah satu link utama, Gi0/3, dimatikan dengan perintah `shutdown`. Koneksi tetap tersedia melalui link Cisco–Huawei yang masih aktif.

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.205 ms  1.889 ms  2.103 ms
 2  10.10.11.2     3.546 ms  1.992 ms  2.673 ms
 3  *192.168.50.10  5.013 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 21: Satu Link Mati

3. Link utama kedua, Gi0/0, kemudian dimatikan. OSPF menghitung ulang jalur dan memindahkan trafik ke jalur cadangan: PC VLAN → Cisco → MikroTik → Huawei → LAN Huawei. Hop 10.10.30.2 dan 10.10.20.1 pada hasil trace membuktikan trafik melewati MikroTik.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.008 ms  2.293 ms  1.523 ms
 2  10.10.30.2     2.880 ms  2.019 ms  1.656 ms
 3  10.10.20.1     4.773 ms  3.144 ms  3.229 ms
 4  *192.168.50.10 6.816 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 22: Dua link utama mati

4. Kedua link Cisco–Huawei diaktifkan kembali menggunakan `no shutdown` sehingga route kembali memakai jalur utama dengan cost lebih kecil.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.20.1    2.344 ms  6.521 ms  1.836 ms
 2  10.10.10.2     3.347 ms  2.813 ms  2.926 ms
 3  *192.168.50.10 3.669 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 23: Kondisi Awal

2 Analisis Hasil Percobaan

Dari percobaan yang dilakukan, pembagian jaringan menggunakan VLAN sudah berjalan sesuai konfigurasi. VLAN 10, 20, 30, dan 40 digunakan untuk memisahkan tiap bagian, yaitu Finance, HR, IT, dan Marketing. Port yang terhubung ke PC juga sudah dimasukkan ke VLAN masing-masing melalui konfigurasi access port. Setelah itu, trunk pada Gi0/0 digunakan agar semua VLAN tersebut bisa lewat menuju Cisco Router.

Pada Cisco Router, subinterface Gi0/2 dibuat untuk setiap VLAN. Subinterface ini digunakan sebagai gateway, sehingga PC dari VLAN yang berbeda tetap bisa saling terhubung. Selain itu, DHCP untuk VLAN 10 dan VLAN 20 juga sudah berjalan karena client mendapatkan alamat IP secara otomatis, yaitu 192.168.10.11 dan 192.168.20.11.

Untuk bagian antar-router, link antara Cisco, Huawei, dan MikroTik sudah dikonfigurasi dan dapat saling ping. Setelah OSPF dijalankan, router-router tersebut mulai saling mengenali network yang tidak terhubung langsung. Hal ini dapat dilihat dari tabel routing dan hasil pengujian ping dari jaringan VLAN Cisco ke LAN Huawei.

Pada pengujian cost OSPF, jalur Cisco–Huawei dipilih sebagai jalur utama karena cost-nya lebih kecil. Sementara itu, jalur melalui MikroTik digunakan sebagai jalur cadangan. Saat dua link utama Cisco–Huawei dimatikan, jalur berpindah melewati MikroTik. Hal ini terlihat dari hasil trace yang melewati 10.10.30.2 dan 10.10.20.1. Setelah link utama dinyalakan kembali, jalur kembali menggunakan koneksi langsung Cisco–Huawei.

3 Hasil Tugas Modul

Hasil tugas modul terdapat pada readme kelompok :

<https://github.com/haikalramadhanasa/-Praktikum-Jarkom-2026-Kelompok-14/tree/main/Modul-5/LA>

4 Kesimpulan

Berdasarkan percobaan yang telah dilakukan, VLAN dan access port berhasil memisahkan empat jaringan divisi, yaitu Finance, HR, IT, dan Marketing. Konfigurasi trunk juga berhasil membawa seluruh VLAN melalui satu link fisik menuju router. Router-on-a-Stick pada Cisco Router berhasil menyediakan gateway untuk setiap VLAN sehingga komunikasi antar-VLAN dapat dilakukan. Selain itu, DHCP pada Cisco Router juga berhasil melayani client pada VLAN 10 dan VLAN 20. Link antara Cisco, Huawei, dan MikroTik berhasil dikonfigurasi sesuai dengan addressing percobaan dan dapat saling berkomunikasi. OSPF multivendor juga berhasil membentuk adjacency serta mendistribusikan route antar-network pada perangkat Cisco, Huawei, dan MikroTik. Pengaturan OSPF cost berhasil menentukan jalur utama dan jalur cadangan. Ketika dua link Cisco–Huawei dimatikan, trafik berpindah melalui MikroTik dan kembali ke jalur utama setelah link dipulihkan.

5 Lampiran

5.1 Dokumentasi Saat Praktikum



Gambar 24: Dokumentasi saat praktikum.